

INTRODUCTION

Applied Dental Biomaterial

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Restorative Workflow



Examination



Photos and X-rays



Diagnosis



Planning and
Material Selection



Cavity preparation
(classes)



**Application Of
Material**



Finishing



EXAMINATION

Photos and X-rays



A close-up photograph of a dentist in a white lab coat and blue gloves examining a patient's teeth. The dentist is using a dental mirror and a probe. The patient is a young woman with blonde hair, lying back in a dental chair. The word "DIAGNOSIS" is overlaid in large white letters across the center of the image.

DIAGNOSIS

The background of the slide is a photograph of a workspace. It features a spiral-bound notebook with several hand-drawn diagrams, likely user interface wireframes, showing various rectangular boxes and flow lines. A hand is visible in the bottom right corner, holding a pen and pointing at one of the diagrams. In the upper right, a portion of a computer keyboard is visible. The entire image is overlaid with a dark blue semi-transparent filter.

PLANNING AND MATERIAL SELECTION



OUTDATED

MATERIAL SELECTION

amalgam



AMALGAMATOR



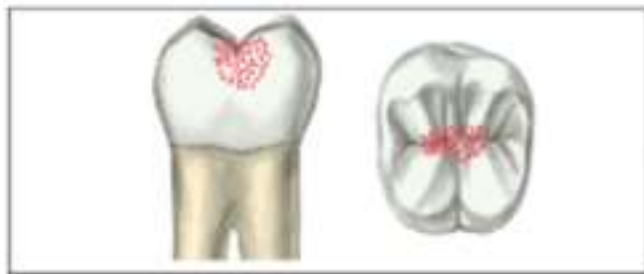
COMPOSITE



GLASS IONOMER



GLASS IONOMER CEMENT



CLASS I



CLASS II



CLASS III



CLASS IV



CLASS V



CLASS VI

CLASSES



CLASS !

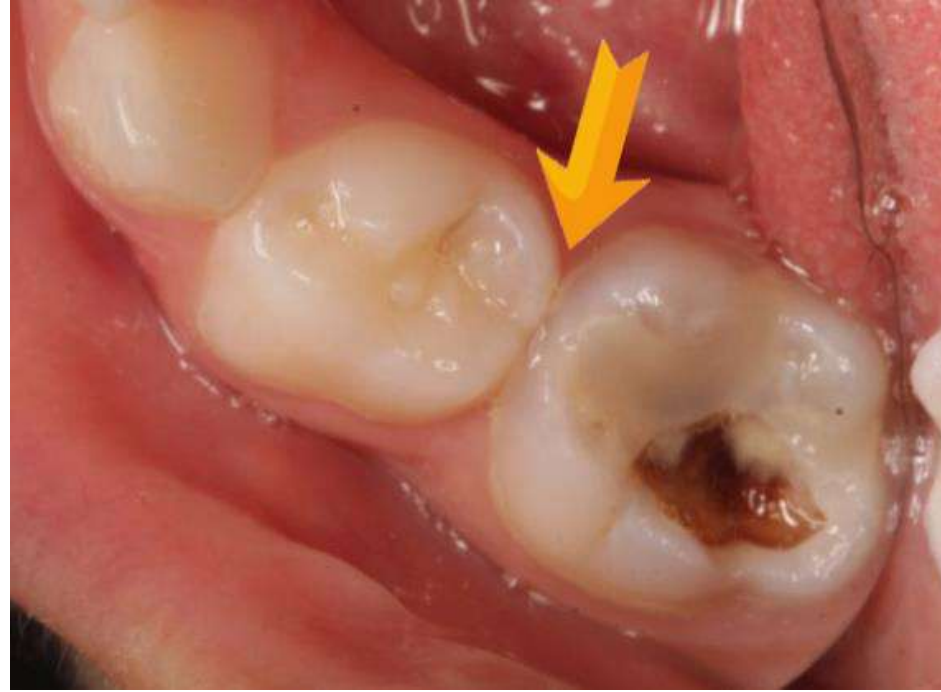
MATERIAL OF SELECTION !?





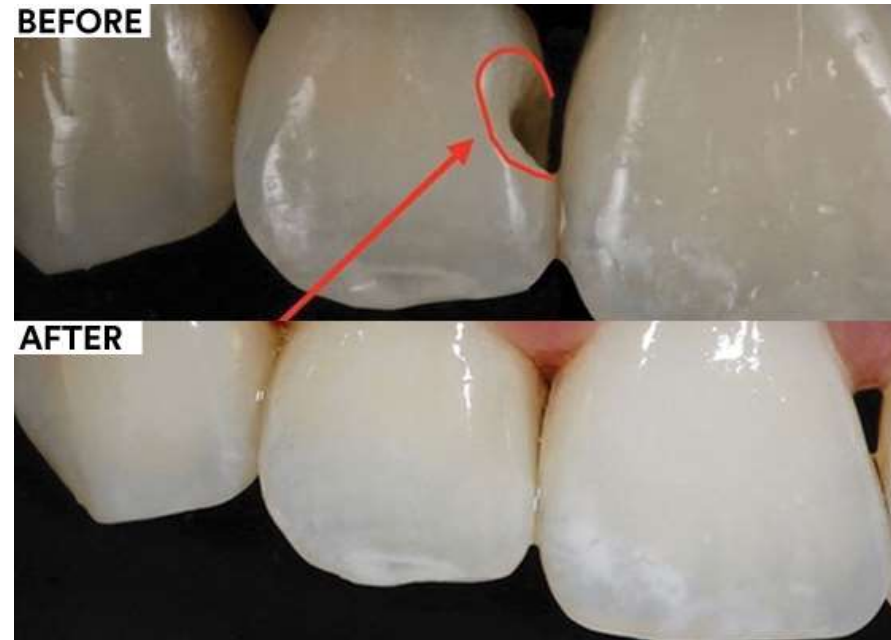
CLASS !

MATERIAL OF SELECTION !



CLASS !?

MATERIAL OF SELECTION !?



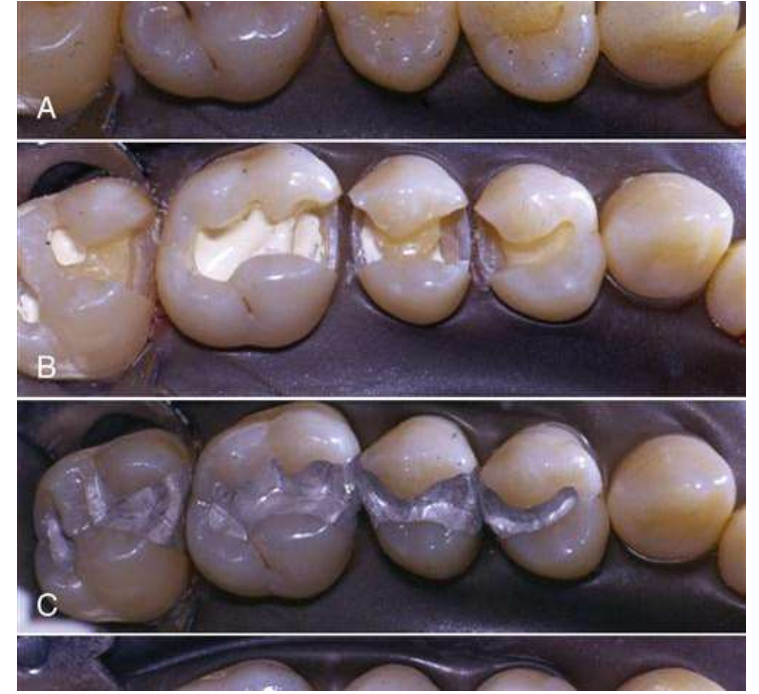
CLASS !?

MATERIAL OF SELECTION !?



CLASS !?

MATERIAL OF SELECTION !?



APPLICATION OF MATERIAL

finishing





ARTICULATING PAPER



Prosthodontics Workflow

Examination

Photos and X-rays

Impression

- Primary impression
- Secondary impression
- Impression material
- Impression trays
- Digital impression
- Impression techniques

Casts

Primary cast (diagnostic cast)

Master cast

Working die materials and techniques of fabrication

Prosthodontics Workflow

Mounting and bite registration

- **Bite registration material**
- Types of articulators
- Digital articulation

Diagnosis

Planning

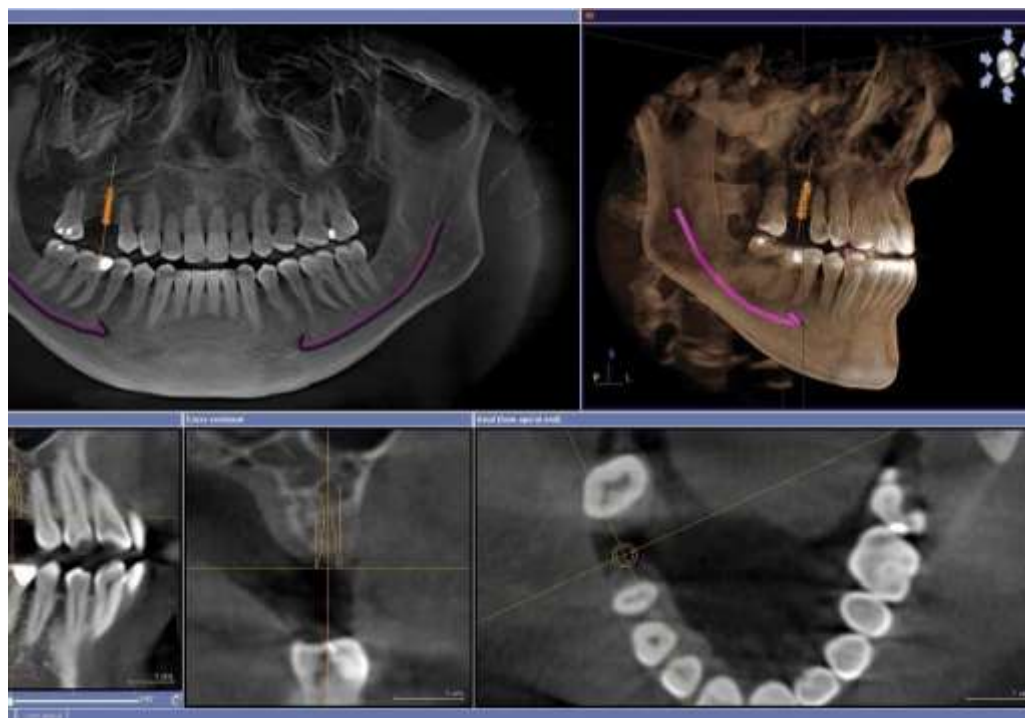
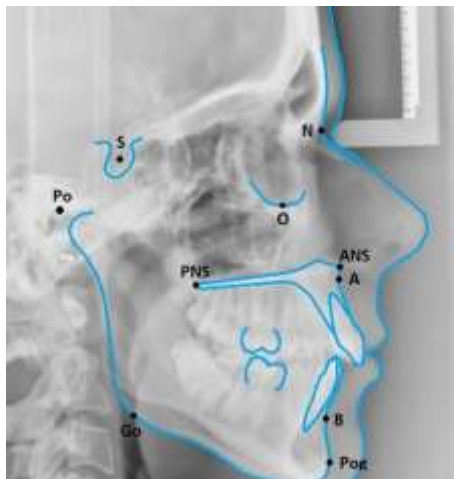
Procedure

- Crown (shape ,material specifications and material of choice)
- Bridge
- Implant
- Complete denture
- Inlay
- Onlay
- Overlay
- Labial veneer
- Occlusal veneer
- Vonlay
- Endo crown

Prosthodontics Workflow

Examination





PROSTHODONTICS WORKFLOW PHOTOS AND X- RAYS

PROSTHODONTICS WORKFLOW IMPRESSION



Types of impression tray...

- I. **Stock Trays** (Various Size & Made By Manufactures)
- II. **Custom** (Individual Or Special) **Trays**
Made For One Patient And Discarded Later



Stock Trays



Custom Tray

Stock Impression tray and construction of diagnostic
By:Dr.Tushar

Impression trays



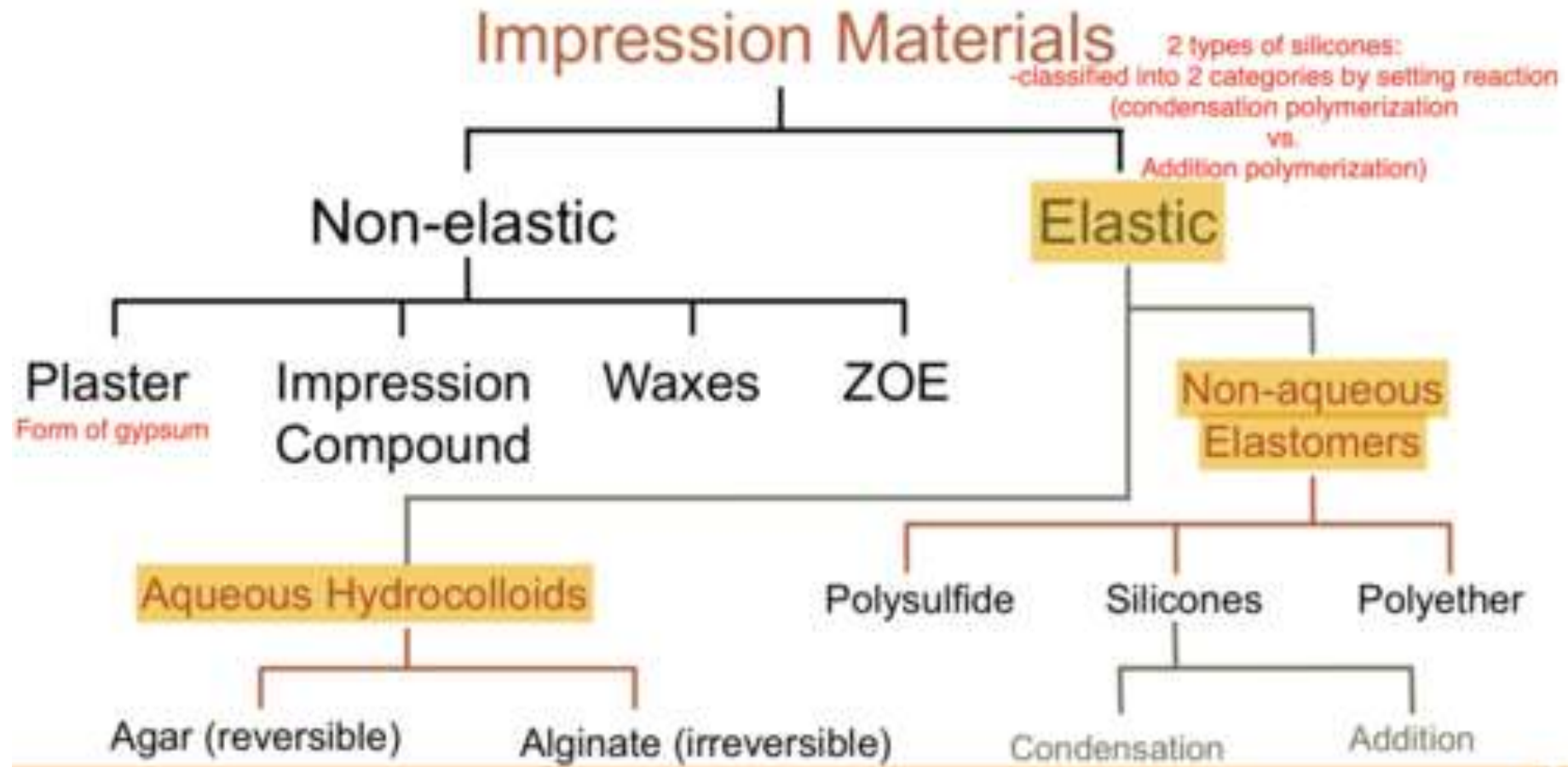
Impression trays



Impression materials



Impression materials



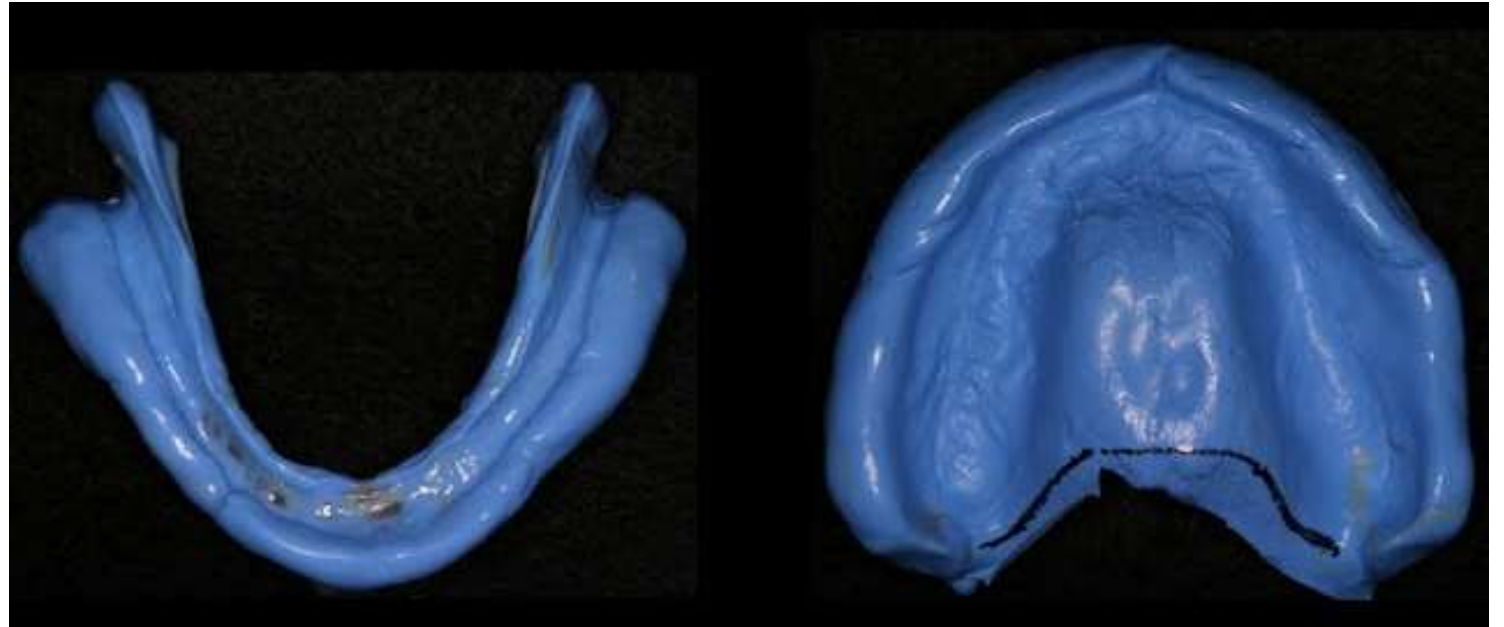


PRIMARY IMPRESSION

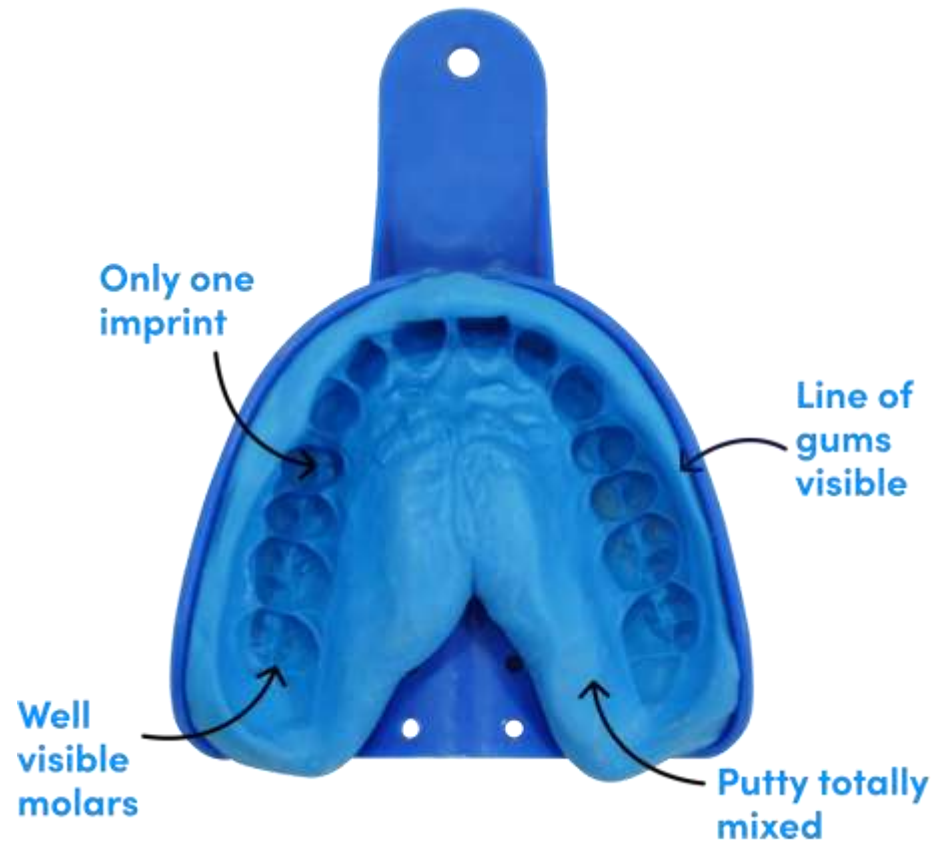


PRIMARY CAST (DIAGNOSTIC CAST)

Secondary impression



Secondary impression



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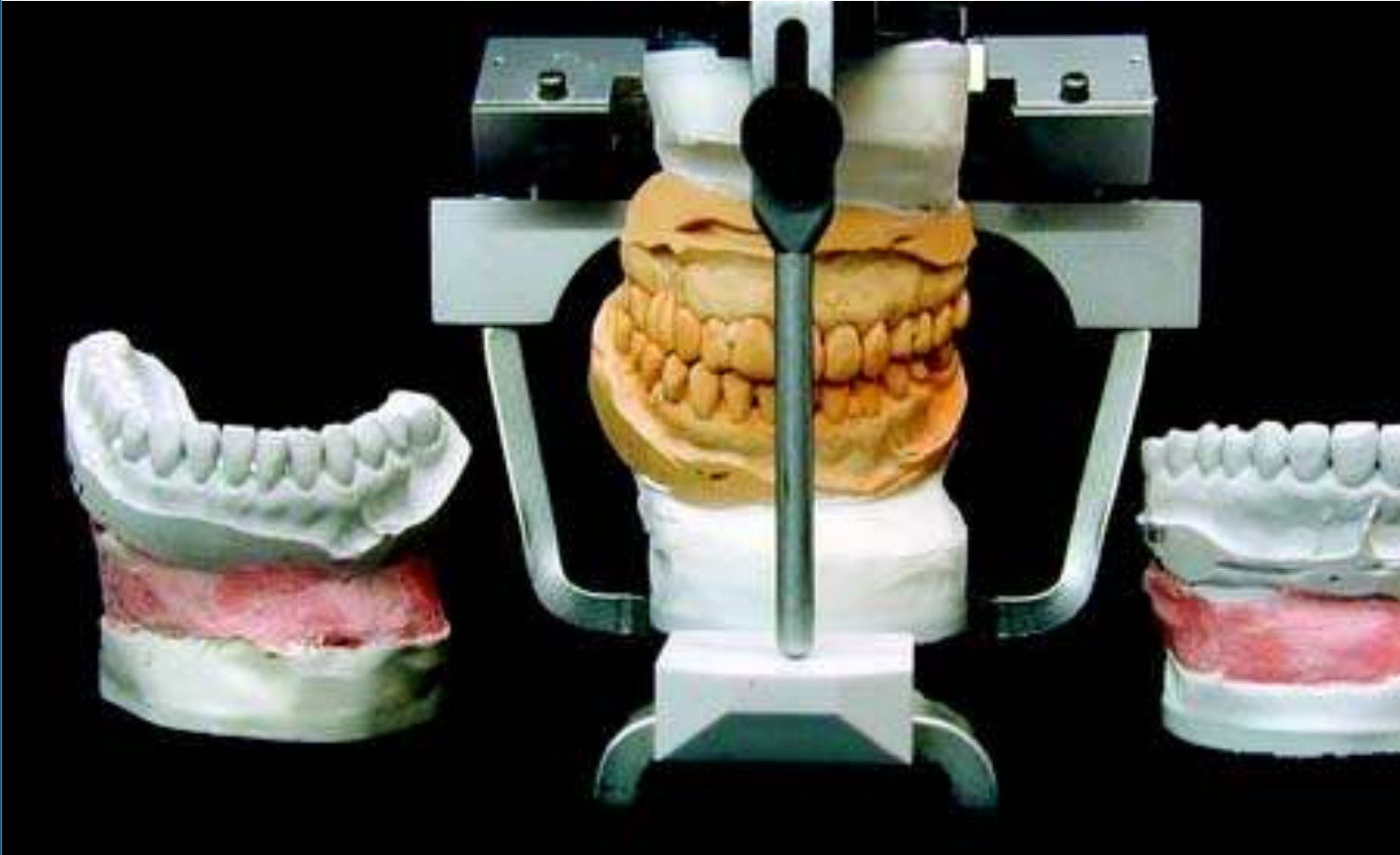
MASTER CAST



WORKING DIE MATERIALS AND TECHNIQUES OF FABRICATION



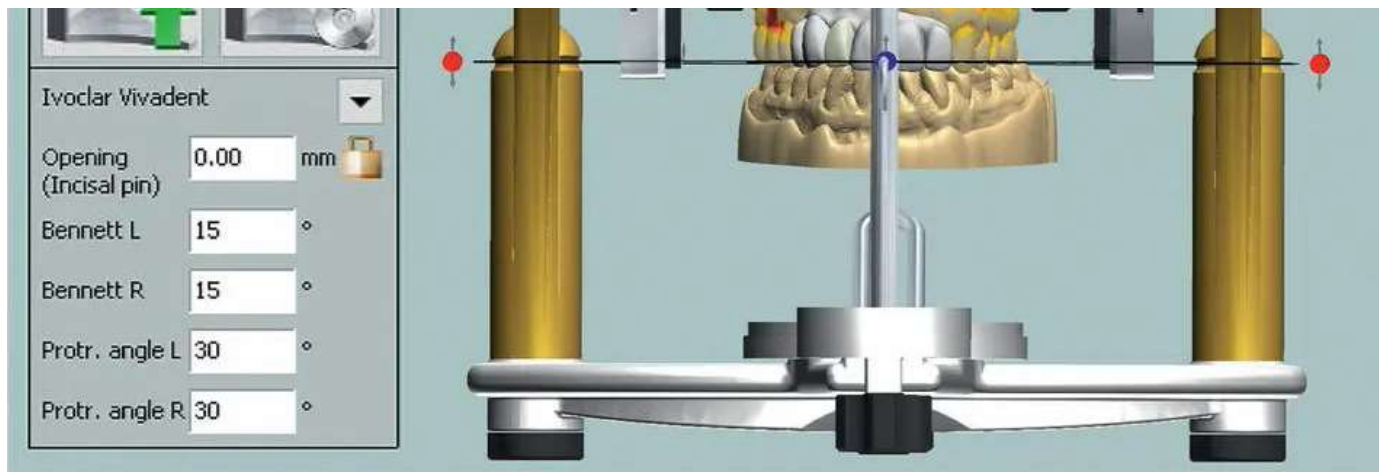
DIGITAL IMPRESSION



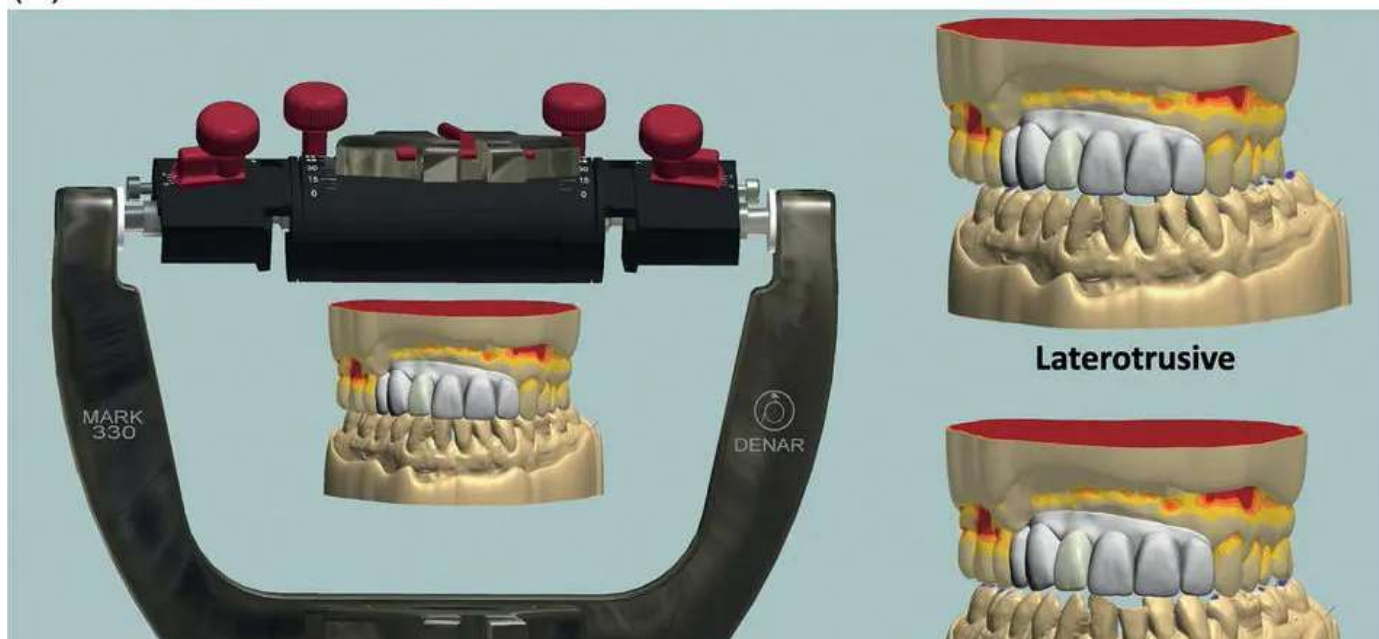
MOUNTING AND BITE REGISTRATION



TYPES OF ARTICULATORS



(b)



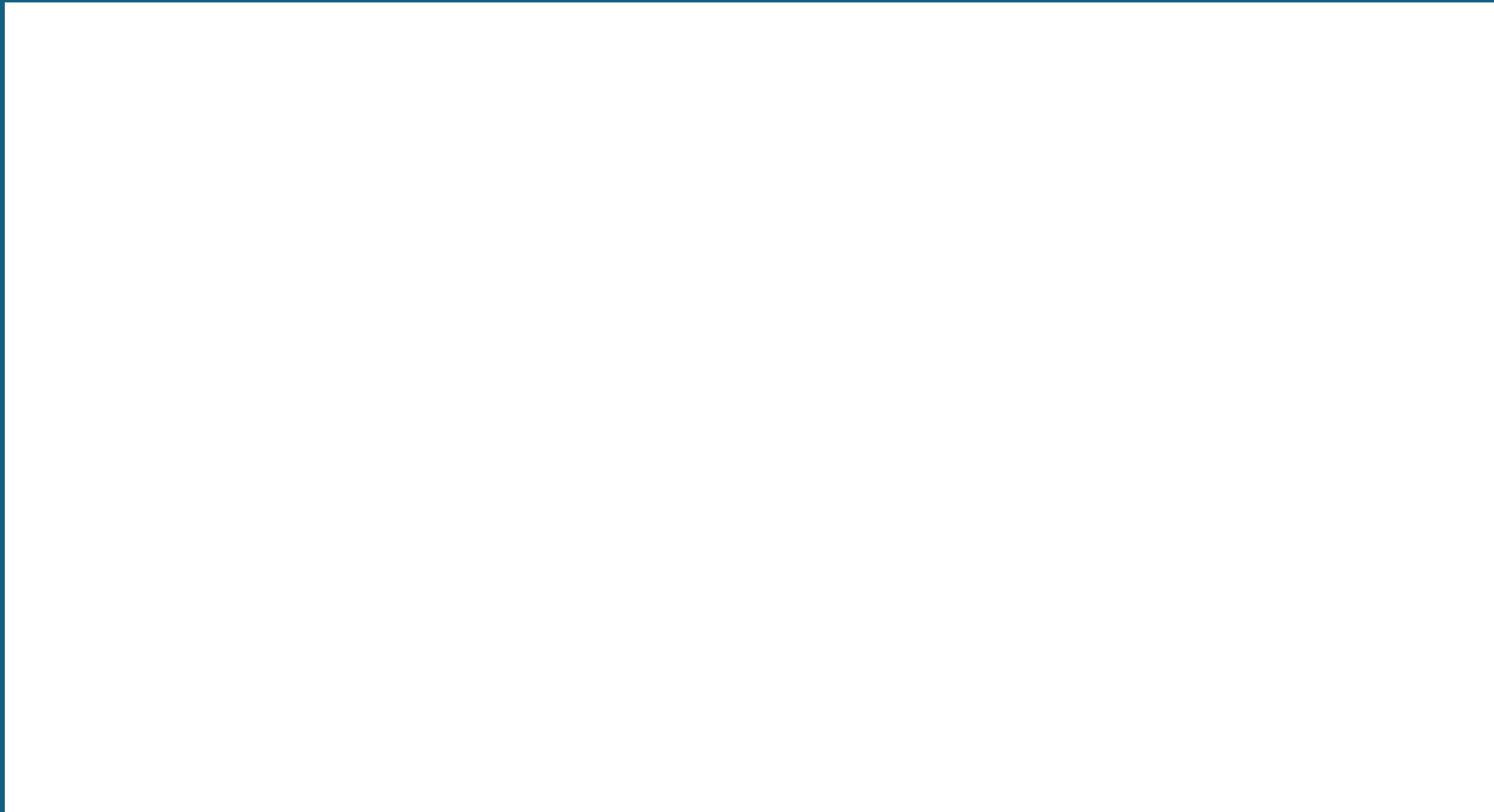
DIGITAL ARTICULATION



BITE REGISTRATION MATERIAL



- complete
denture
video



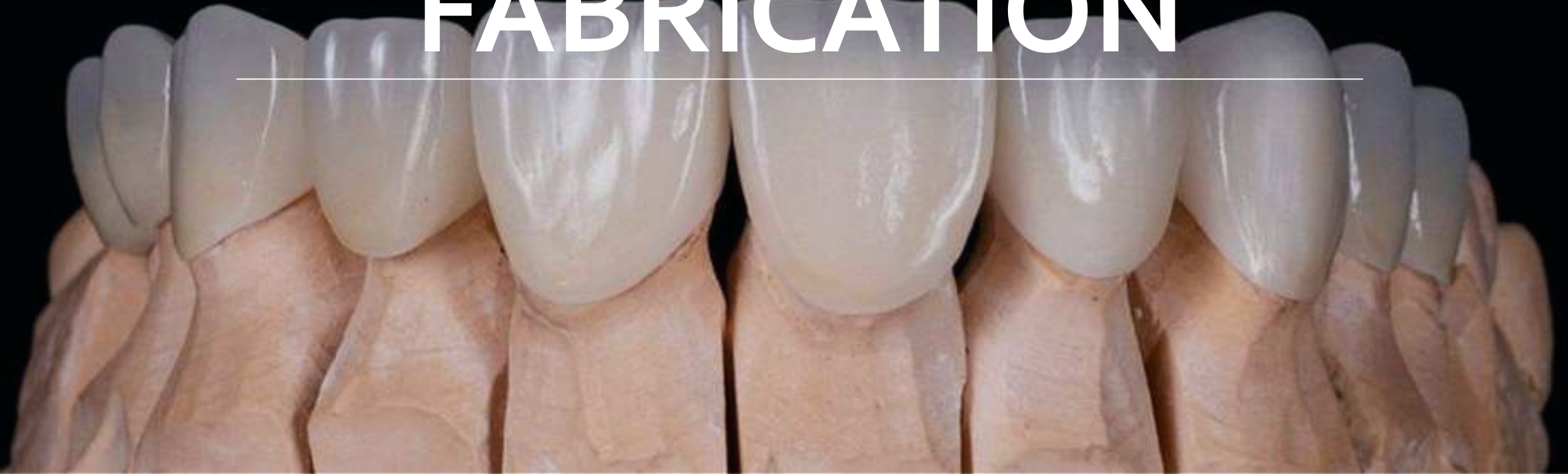
DIAGNOSIS

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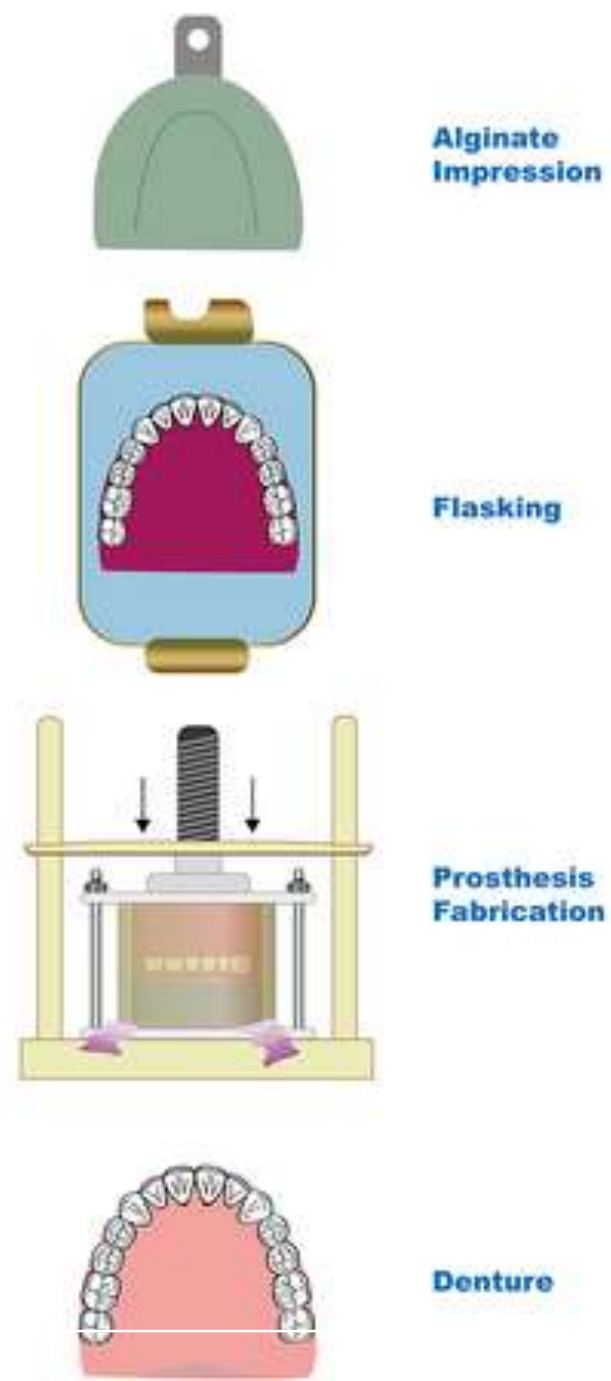
A dental professional with long brown hair, wearing blue scrubs, is seated and holding a small, pinkish-red dental model of a tooth. She is looking down at the model with a focused expression. A patient with long, wavy brown hair, wearing a white shirt, is seated in a dental chair, facing the professional. The background shows a clinical setting with various dental equipment, including a large white machine with multiple nozzles. The word "PLANNING" is overlaid in large, white, sans-serif capital letters across the center of the image.

PLANNING

FABRICATION

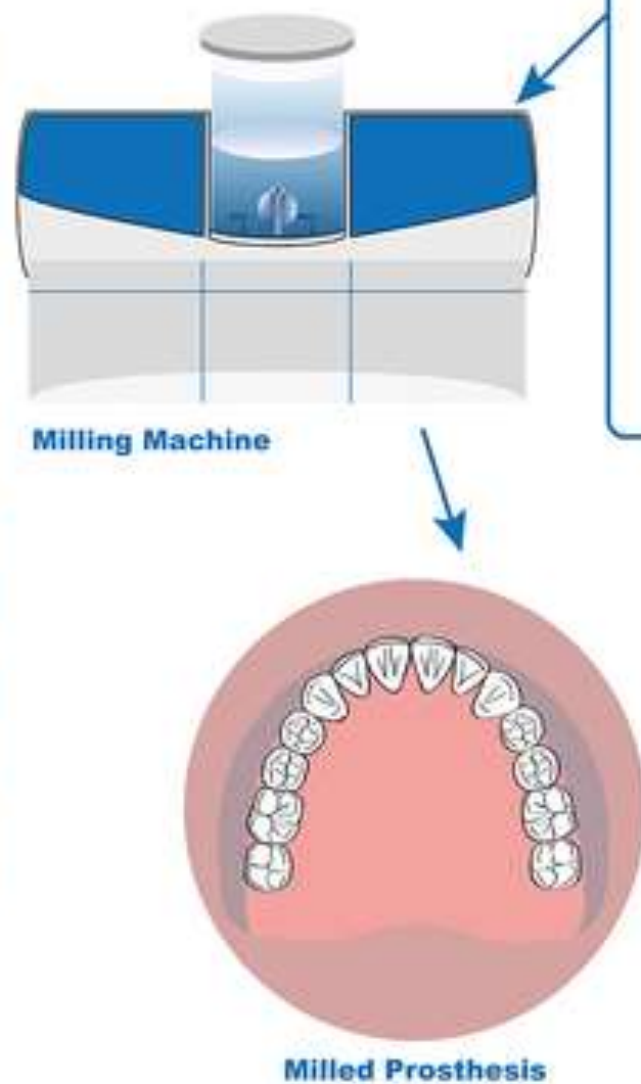


Formative Manufacturing

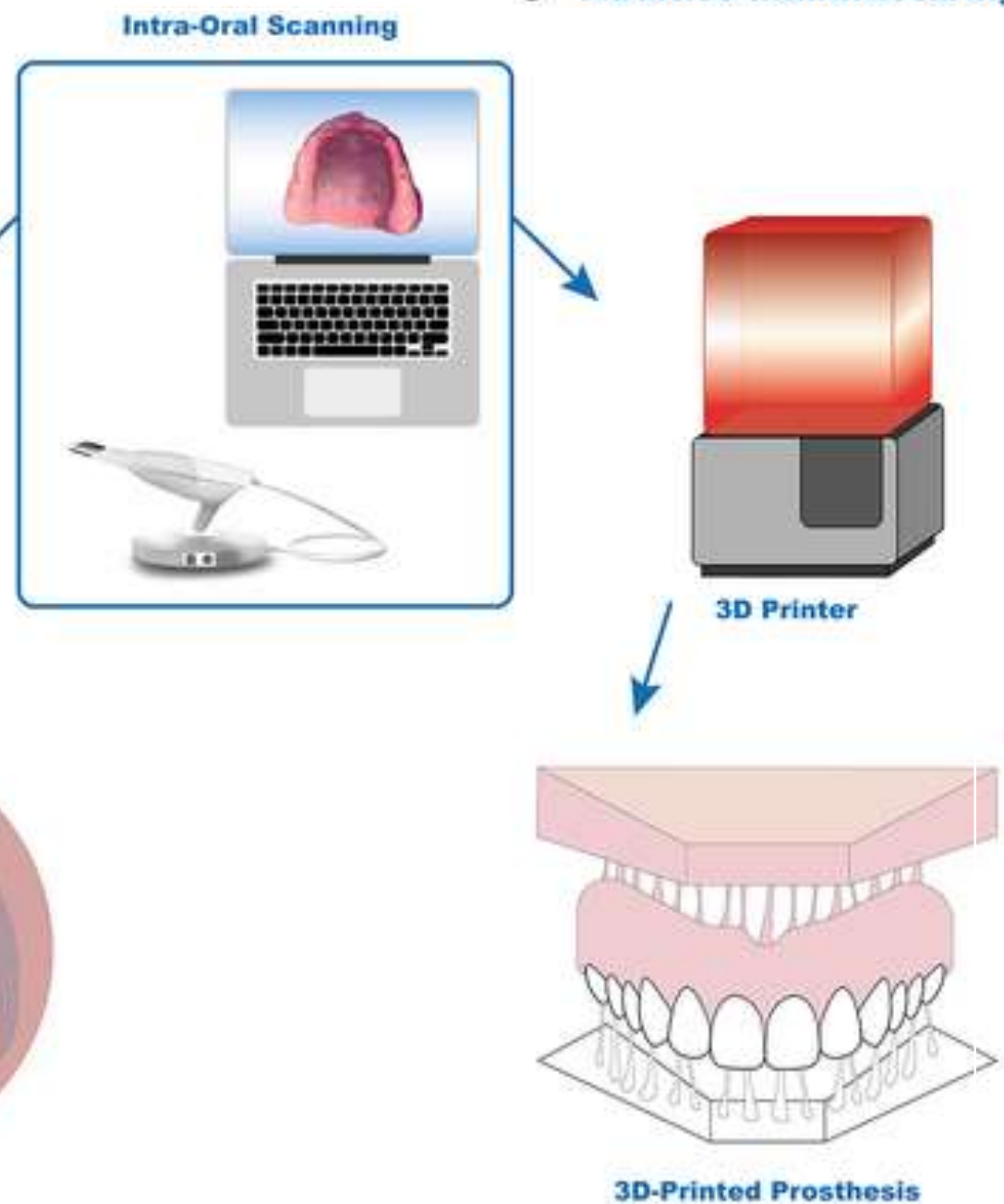


B Subtractive Manufacturing

DIGITAL DENTURE PROCESSING



C Additive Manufacturing



Prosthesis

Removable

- Complete
- Partial

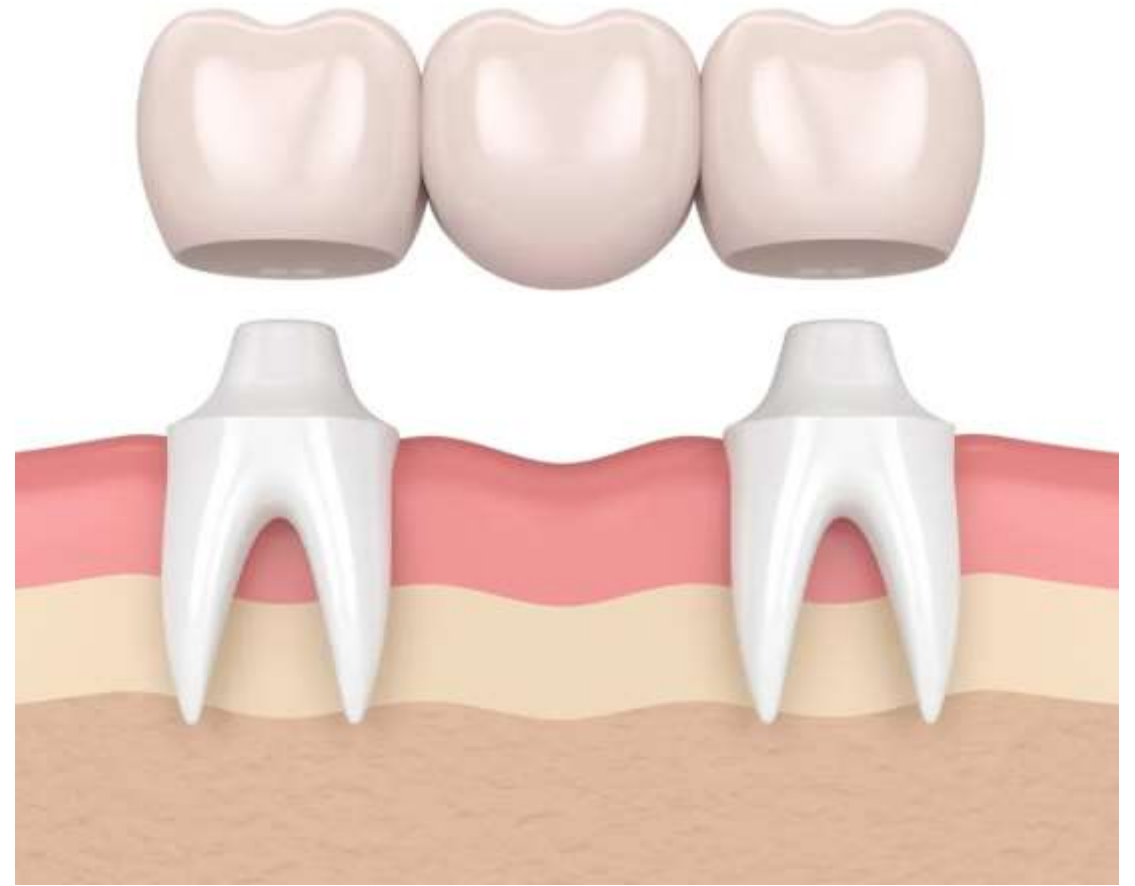
Fixed

- Crown
- Bridge

Crown

- PFM
- Ceramics
- Metal







BRIDGE



BRIDGE



BRIDGE



IMPLANT

TITANIUM



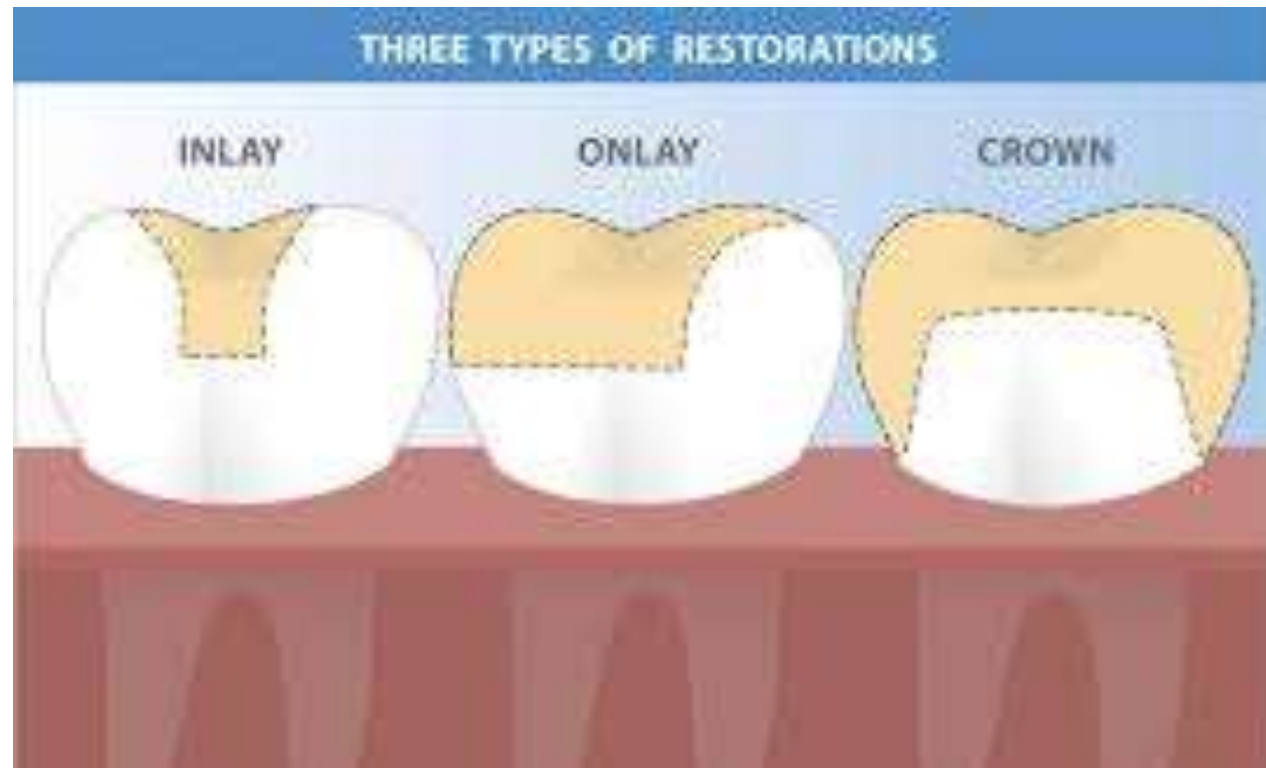
PARTIAL DENTURE



COMPLETE DENTURE

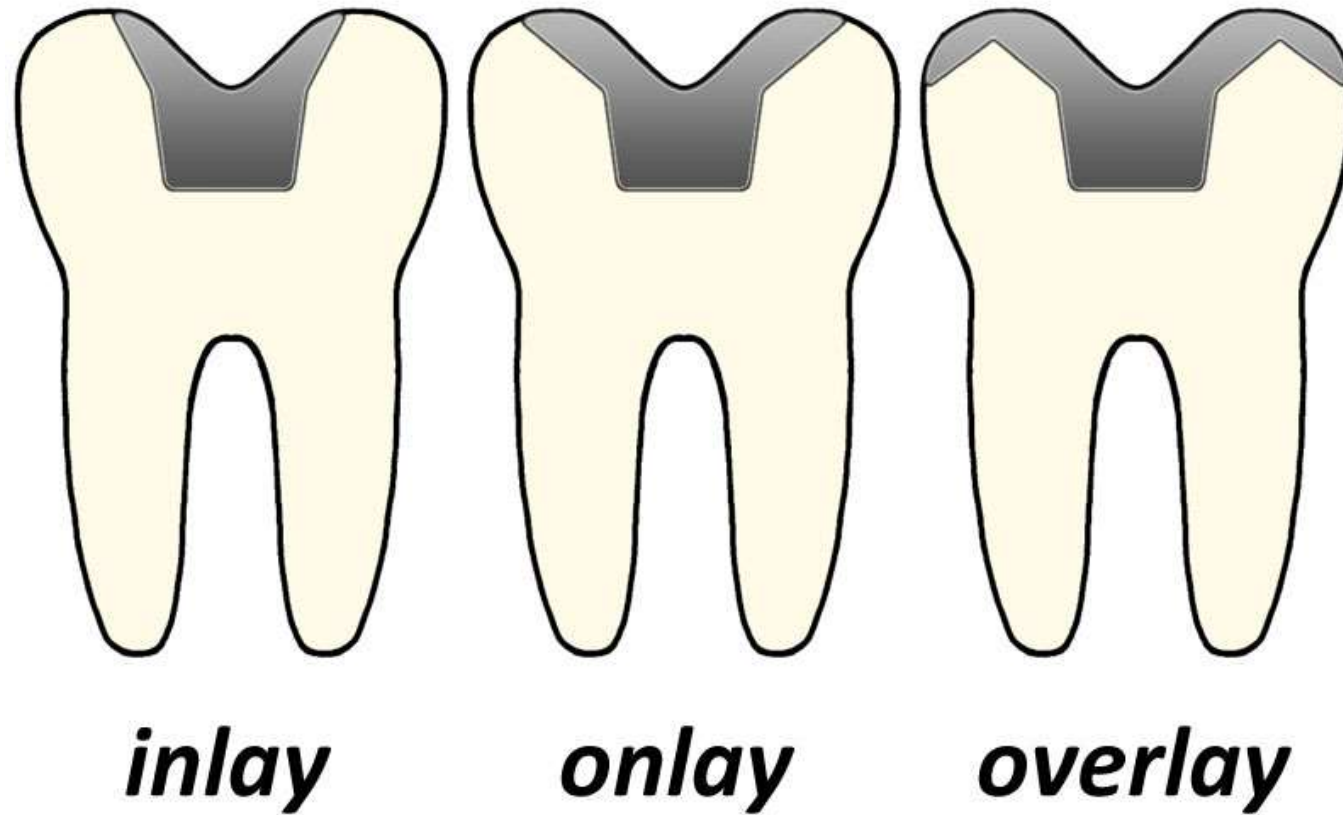
HEAT CURE ACRYLIC RESIN

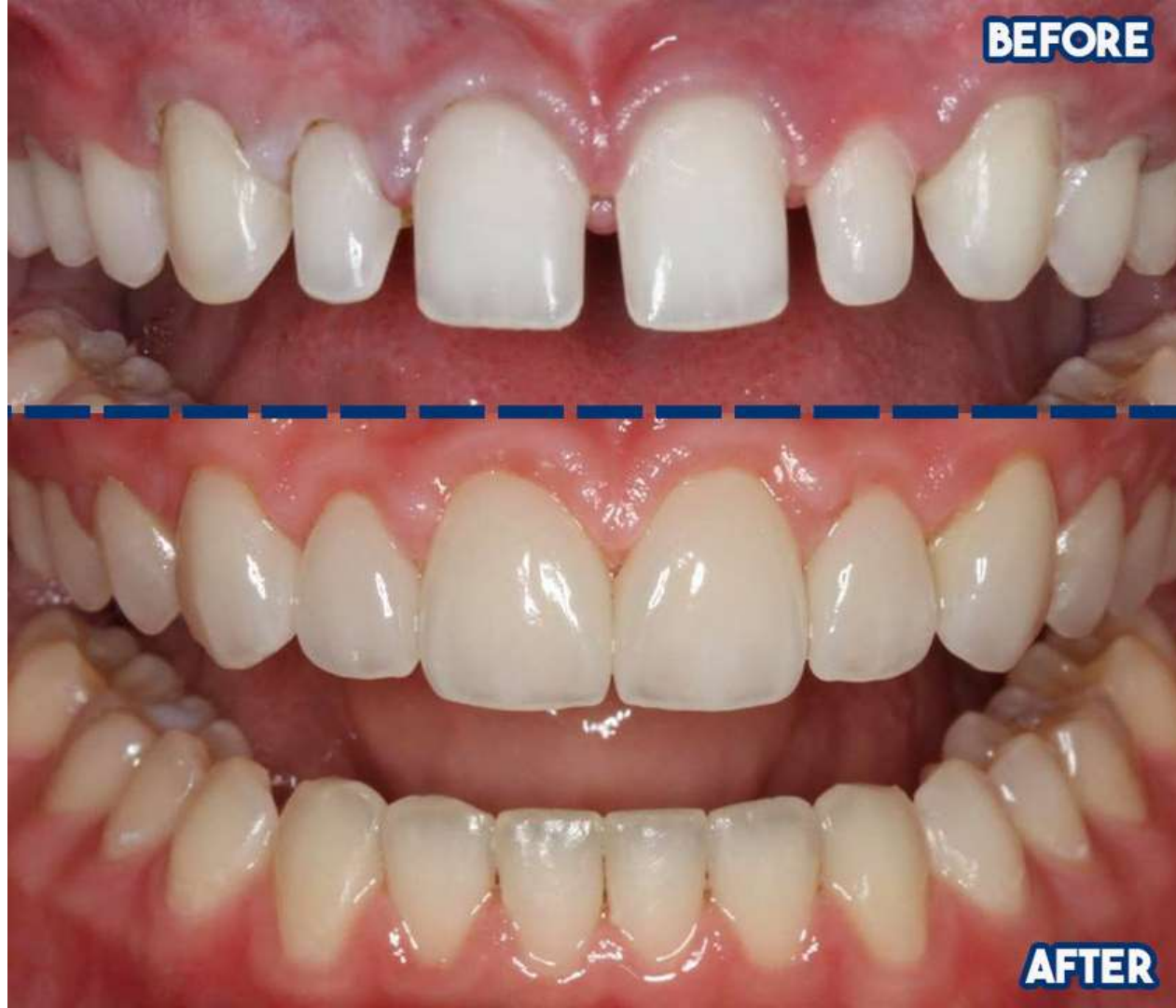
INLAY



- Gold
- Metal
- Ceramics
- Hybrid ceramics

Indirect restoration





**LABIAL
VENEER**

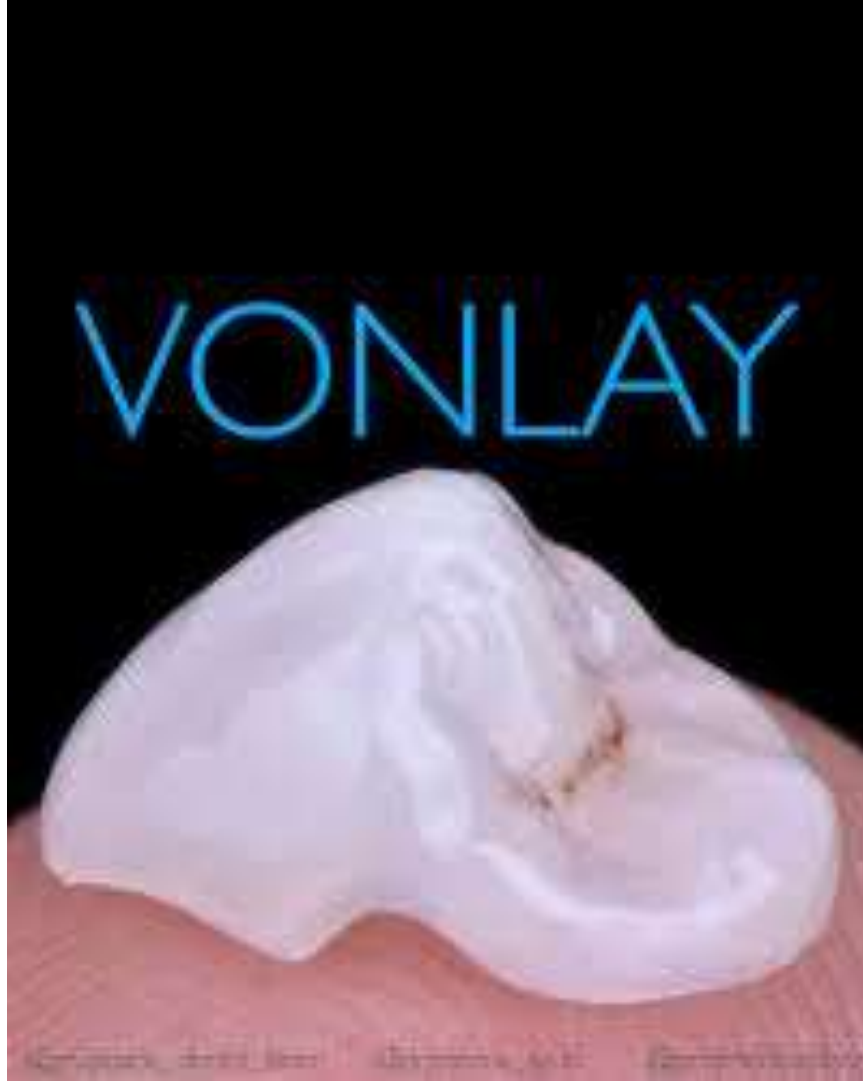
**LITHIUM
DISILICATE**



OCCLUSAL VENEER



Endo crown



VONLAY

